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A. Initial data exploration

A1. Attribute Types

S/N	Attribute	Type	Justification
1	Date	Ordinal	Dates are sequential (ordered), but do not have a true zero point, and intervals cannot be meaningfully compared.
2	Location	Nominal	Locations are categorical, with no inherent order or magnitude.
3	MinTemp	Interval	Temperature is measured on an interval scale, with consistent intervals and no true zero (negative values exist).
4	MaxTemp	Interval	Same as MinTemp, measured on an interval scale.
5	Rainfall	Ratio	Rainfall has a true zero point (no rainfall) and can be compared proportionally, making it a ratio attribute.
6	Evaporation	Ratio	Evaporation is measured with a true zero point, allowing for proportional comparison between amounts.
7	Sunshine	Ratio	Hours of sunshine have a true zero (no sunshine) and are measured on a ratio scale, allowing for comparison.
8	WindGustDir	Nominal	Wind direction is a categorical variable with no inherent order (e.g., N, S, E, W).
9	WindGustSpeed	Ratio	Wind speed has a meaningful zero point (no wind), and the intervals and ratios can be compared.
10	WindDir9am	Nominal	Wind direction at 9am is categorical and does not have any inherent order.
11	WindDir3pm	Nominal	Same as WindDir9am, categorical with no intrinsic order.
12	WindSpeed9am	Ratio	Wind speed is measured on a ratio scale with a true zero (no wind).
13	WindSpeed3pm	Ratio	Same as WindSpeed9am, it is measured on a ratio scale.
14	Humidity9am	Ratio	Humidity, expressed as a percentage (0%

			representing no humidity), is a ratio attribute.
15	Humidity3pm	Ratio	Same as Humidity9am.
16	Pressure9am	Ratio	Atmospheric pressure has a true zero (theoretically), making it a ratio attribute.
17	Pressure3pm	Ratio	Same as Pressure9am
18	Cloud9am	Ratio	Cloud cover is measured in oktas (0 to 8), where 0 represents no clouds, making it a ratio attribute as it has a true zero
19	Cloud3pm	Ratio	Same as Cloud9am, measured in oktas, with a true zero point and meaningful proportional comparison.
20	Temp9am	Interval	Temperature at 9am is an interval variable, as with MinTemp and MaxTemp, because it lacks a true zero.
21	Temp3pm	Interval	Same as Temp9am.
22	RainToday	Nominal	A categorical variable indicating if it rained or not (yes/no), making it nominal.
23	RainTomorrow	Nominal	Similar to RainToday, this is a categorical variable that likely indicates if rain is forecasted or not.

A2. The summarising properties for the attributes

Frequency & Distribution Data Visualisations

ORDINAL ATTRIBUTES

Attribute: Date

Statistics	Value
Minimum Date	2007-11-25
Maximum Date	2017-06-24
Count of Unique Dates	1795
Frequency Distribution of Dates	2618
Range of Dates	3,497 days
Days between Minimum and Maximum	998
Most Frequent Date	2012-05-07
Count of Weekdays	1891
Count of Weekends	727
Cumulative Frequency of Dates	1795

NOMINAL ATTRIBUTES

Attribute: Location

Statistics	Value
Cumulative Distribution of Locations	2618
Count of Unique Location	49
Most Frequent Location (Mode)	AliceSprings (73)
Least Frequent Location	Uluru (20)
Location with the Maximum Rainfall	Townsville (120.8)
Bar Chart Representation (cross-tab analysis)	Bar Chart Figure 2. Bar Chart for attribute Location This bar chart represents the frequency of occurrences of the main locations.

Attribute: WinGustDir

Statistics	Value
Cumulative Distribution of Directions	2618
Count of Unique Wind Directions	16
Most Frequent Wind Direction (Mode)	SE (369)
Least Frequent Wind Direction	NNW (111)

Destiny Plot Representation (cross-tab analysis)

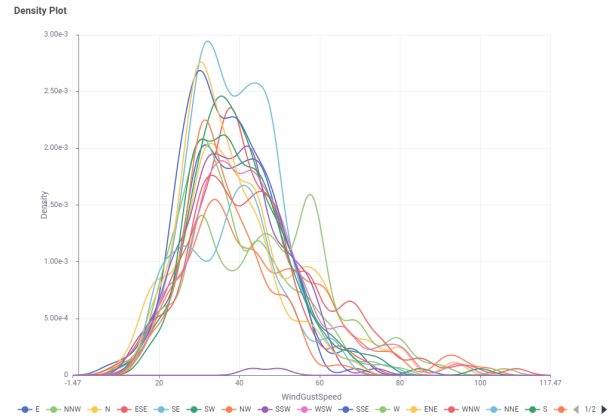


Figure 3. Density Plot for attribute WinGustDir

The density plot for WindGustDir shows the distribution of wind gust directions. SE is the most frequent direction, while NNW is the least common, suggesting directional patterns in wind gusts.

Attribute: WinDir9am

Statistics	Value
Cumulative Distribution of Directions	2618
Count of Unique Wind Directions	16
Most Frequent Wind Direction (Mode)	N
Least Frequent Wind Direction	WSW
Destiny Plot Representation (cross-tab analysis)	Figure 4. Density Plot for attribute WinDir9am This plot highlights the distribution of wind directions at 9 am. The North direction (N) is most common, and WSW is the least

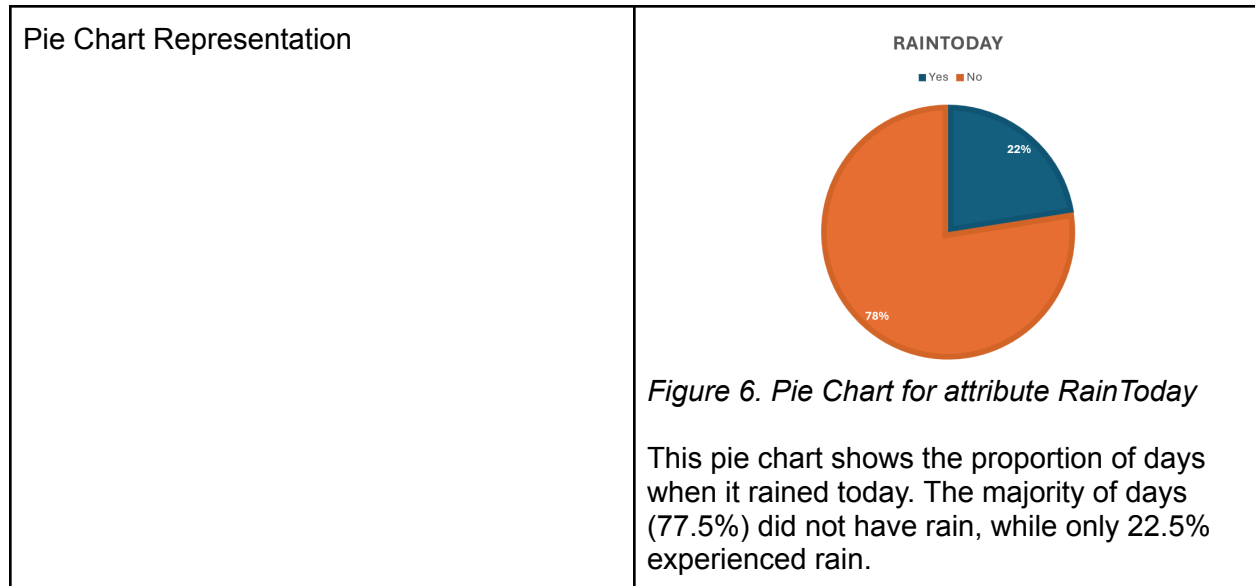
	frequent, indicating a pattern in morning wind directions.
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Attribute: WinDir3pm

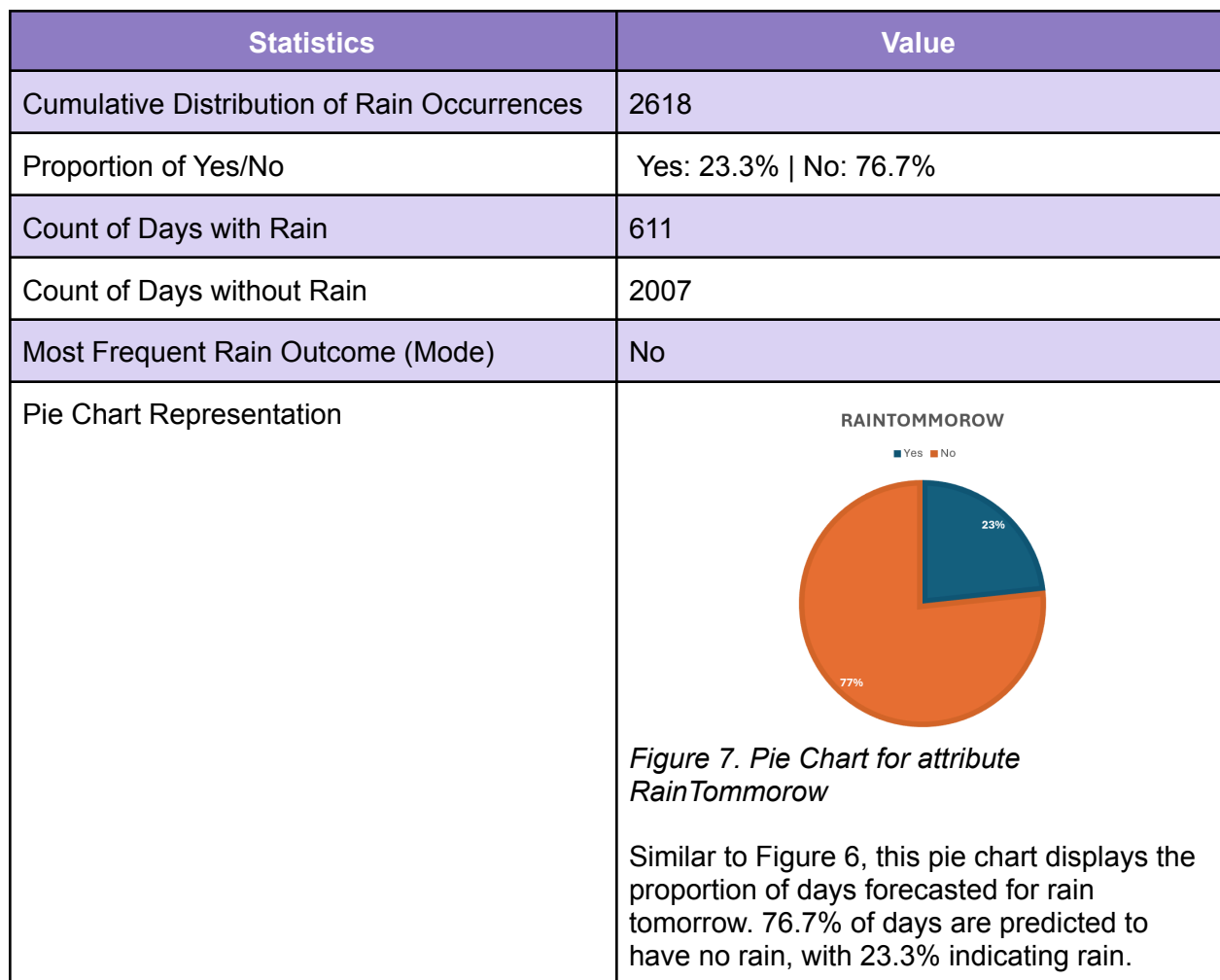
Statistics	Value
Cumulative Distribution of Directions	2618
Count of Unique Wind Directions	16
Most Frequent Wind Direction (Mode)	S
Least Frequent Wind Direction	NNE
Destiny Plot Representation (cross-tab analysis)	<i>Figure 5. Density Plot for attribute WinDir3pm</i> WinDir3pm shows the distribution of wind directions in the afternoon, with South (S) being the most frequent and NNE the least frequent.

Attribute: RainToday

Statistics	Value
Cumulative Distribution of Rain Occurrences	2618
Proportion of Yes/No	Yes: 22.5% No: 77.5%
Count of Days with Rain	588
Count of Days without Rain	2030
Most Frequent Rain Outcome (Mode)	No



Attribute: RainTommorow



INTERVAL ATTRIBUTES

Attribute: MinTemp

Statistics	Value
Minimum Temperature	-6.8
Maximum Temperature	28.8
Mean Temperature	12.137
Median Temperature	12
Mode of Temperature	9.6
Range of Temperatures	35.6
Variance	42.4
Standard Deviation	6.499
25th Percentile (Q1)	7.6
75th Percentile (Q3)	16.8

Histogram

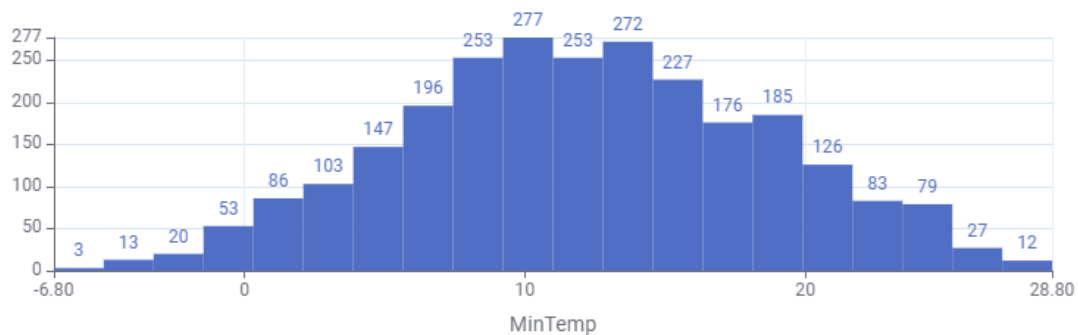


Figure 8. Histogram for attribute MinTemp

This histogram shows the distribution of minimum temperatures. Most temperatures fall between 7.6°C and 16.8°C, with a slight skew towards lower temperatures.

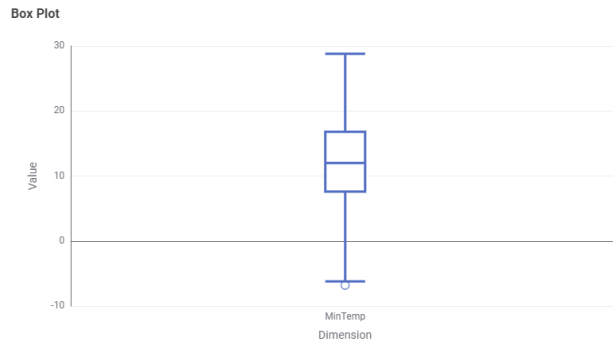


Figure 9. Box Plot for attribute MinTemp

The box plot highlights the spread of minimum temperatures, with a median of 12°C and some outliers at the lower end, indicating unusually cold days.

Attribute: MaxTemp

Statistics	Value
Minimum Temperature	-2.4
Maximum Temperature	46.1
Mean Temperature	23.202
Median Temperature	22.7
Mode of Temperature	18.8
Range of Temperatures	48.5
Variance	52.531
Standard Deviation	7.248
25th Percentile (Q1)	18
75th Percentile (Q3)	28.4

Histogram

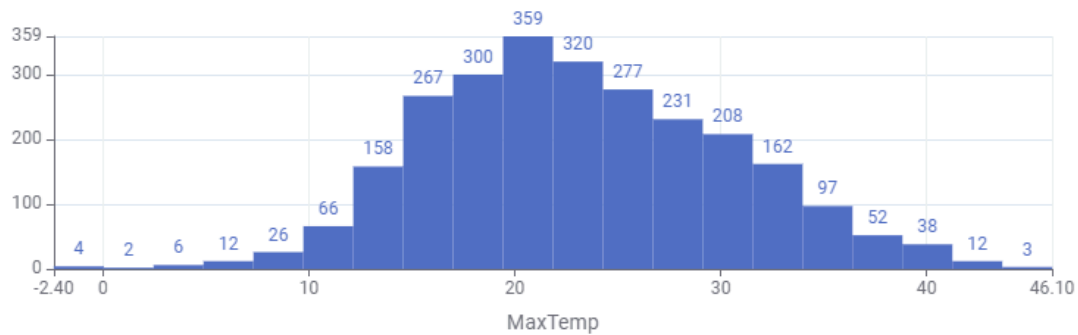


Figure 10. Histogram for attribute MaxTemp

The histogram shows the distribution of maximum temperatures. Most values fall between 18°C and 28.4°C, with the peak at 22.7°C.

Box Plot

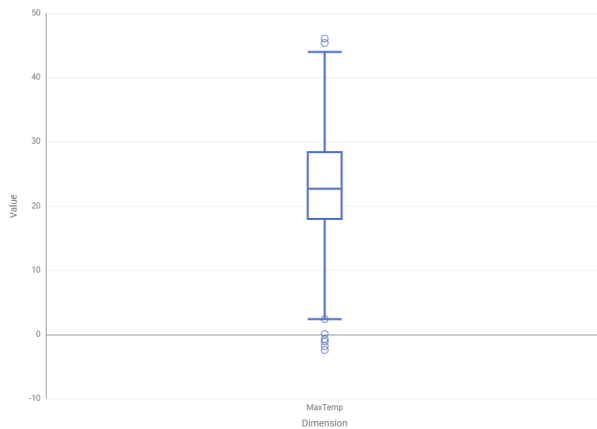


Figure 11. Box Plot for attribute MaxTemp

This box plot shows the distribution of maximum temperatures, with a median of 22.7°C and some outliers at the upper end (above 46.1°C), suggesting extreme hot days.

Attribute: Temp9am

Statistics	Value
Minimum Temperature	-4.2
Maximum Temperature	37.3
Mean Temperature	16.956

Median Temperature	16.8
Mode of Temperature	13.6
Range of Temperatures	41.5
Variance	44.048
Standard Deviation	6.637
25th Percentile (Q1)	12.4
75th Percentile (Q3)	21.5

Histogram

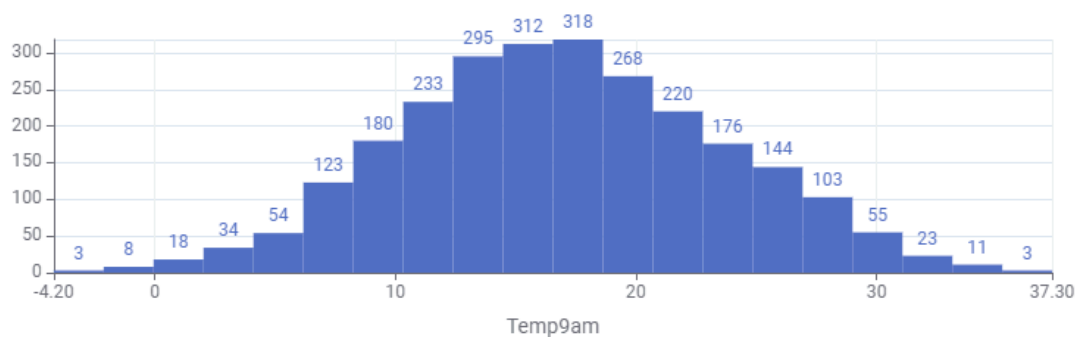


Figure 12. Histogram for attribute Temp9am

The temperature at 9am shows most data points between 12.4°C and 21.5°C, with the peak at around 16.8°C.

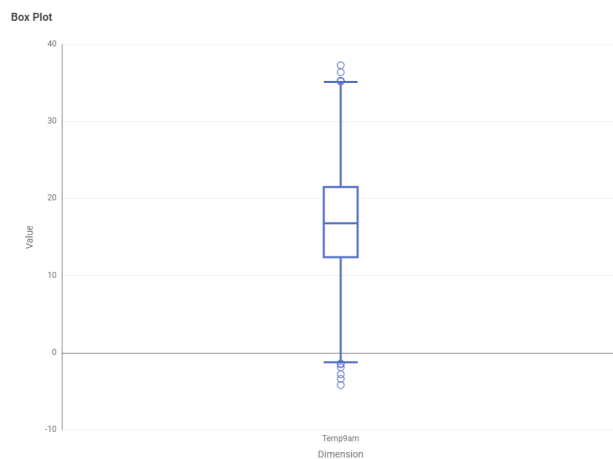


Figure 13. Box Plot for attribute Temp9am

This box plot shows the distribution of temperatures at 9am, with the median at 16.8°C. There are some outliers below -4.2°C, indicating colder mornings.

Attribute: Temp3pm

Statistics	Value
Minimum Temperature	-2.8
Maximum Temperature	44.8
Mean Temperature	21.644
Median Temperature	21.1
Mode of Temperature	22
Range of Temperatures	47.6
Variance	50.644
Standard Deviation	7.116
25th Percentile (Q1)	16.7
75th Percentile (Q3)	26.55

Histogram

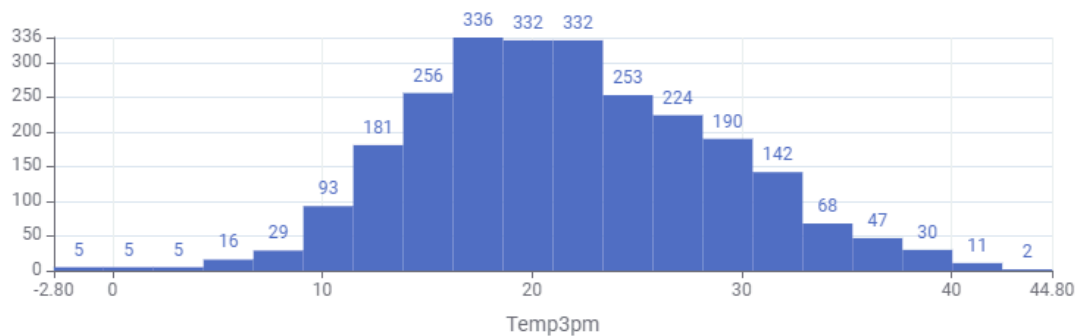


Figure 14. Histogram for attribute Temp3pm

This histogram shows temperatures at 3pm, with a concentration between 16.7°C and 26.55°C, and the highest frequency around 21.1°C.

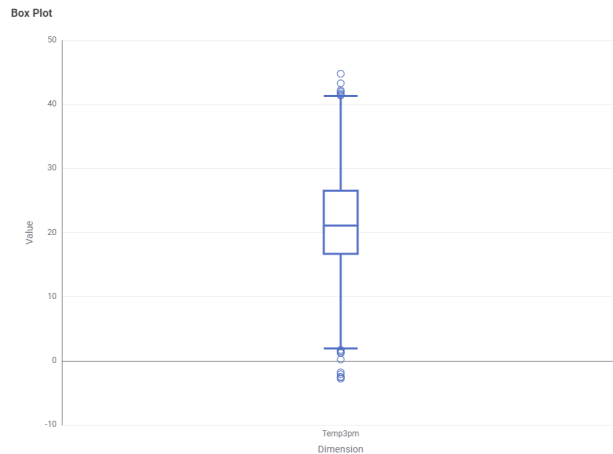


Figure 15. Box Plot for attribute Temp3pm

The box plot displays the spread of 3pm temperatures, with the median at 21.1°C. Outliers on the lower end suggest some unusually cold afternoons

RATIO ATTRIBUTES

I was easily able to access some helpful statistics on the Ratio set of attributes using KNIME's Statistics node. The statistics node is a useful tool for Data Visualization since it graphs a histogram, which shows the distribution and skewness of the data.

Attribute: Rainfall

Statistics	Value
Minimum Rainfall	0
Maximum Rainfall	120.8
Total Rainfall	5749.7
Mean Rainfall	2.241
Median Rainfall	0
Mode of Rainfall	0
Variance	51.225
Standard Deviation	7.157
25th Percentile (Q1)	0
75th Percentile (Q3)	0.8

Density Plot

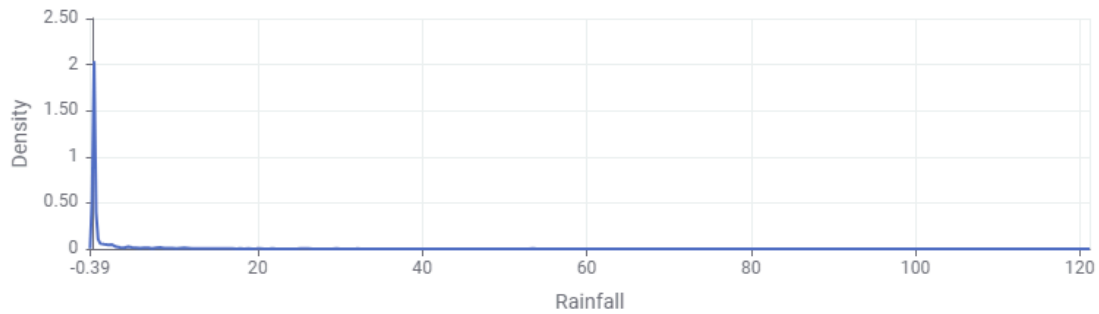


Figure 16. Density Plot for attribute Rainfall

This density plot shows that rainfall is heavily skewed towards 0, meaning many days experienced little to no rainfall.

Box Plot

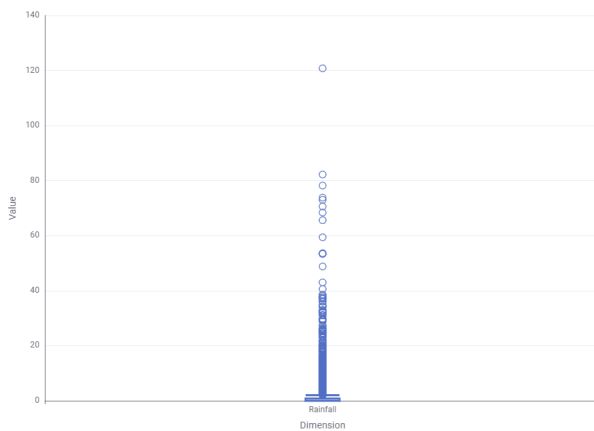


Figure 17. Box Plot for attribute Rainfall

The box plot for rainfall shows that most days have 0 rainfall, with a few extreme outliers indicating heavy rain days (above 120.8 mm).

Attribute: Evaporation

Statistics	Value
Minimum Evaporation	0
Maximum Evaporation	70.4
Total Evaporation	8512.4
Mean Evaporation	5.586

Median Evaporation	4.8
Mode of Evaporation	4
Variance	18.721
Standard Deviation	4.327
25th Percentile (Q1)	2.8
75th Percentile (Q3)	7.4

Density Plot

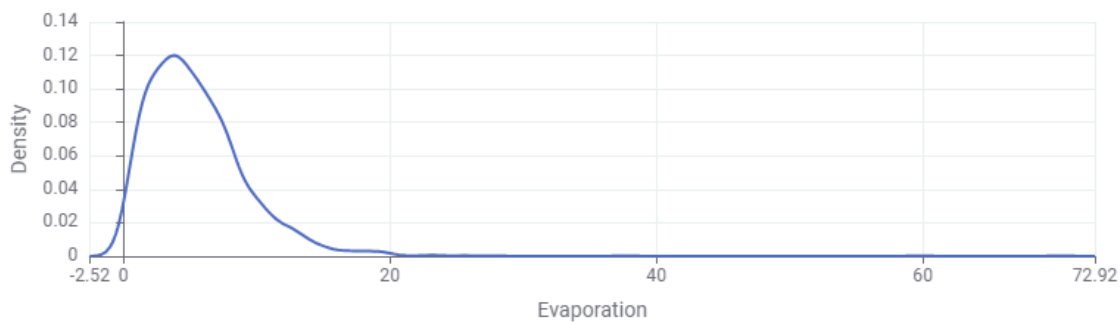


Figure 18. Density Plot for attribute Evaporation

Evaporation follows a similar pattern to rainfall, with most data points close to 0, but some days show higher evaporation rates, peaking at 70.4 mm.

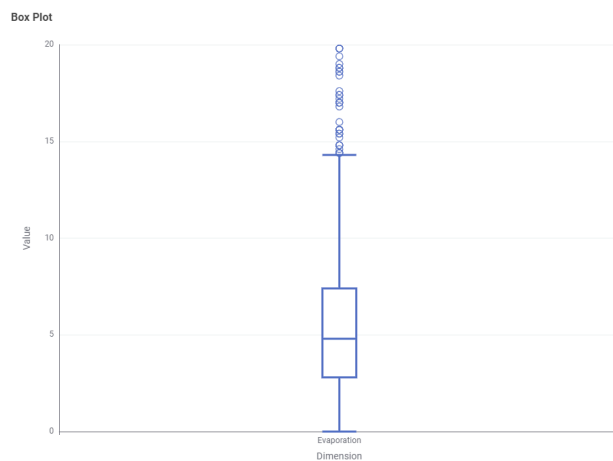


Figure 19. Box Plot for attribute Evaporation

The box plot shows that most evaporation values fall between 2.8 mm and 7.4 mm, with a few outliers for extremely high evaporation.

Attribute: Sunshine

Statistics	Value
Minimum Sunshine	0
Maximum Sunshine	14
Total Sunshine	10747.9
Mean Sunshine	7.694
Median Sunshine	8.7
Mode of Sunshine	0
Variance	14.536
Standard Deviation	3.813
25th Percentile (Q1)	5
75th Percentile (Q3)	10.7

Histogram

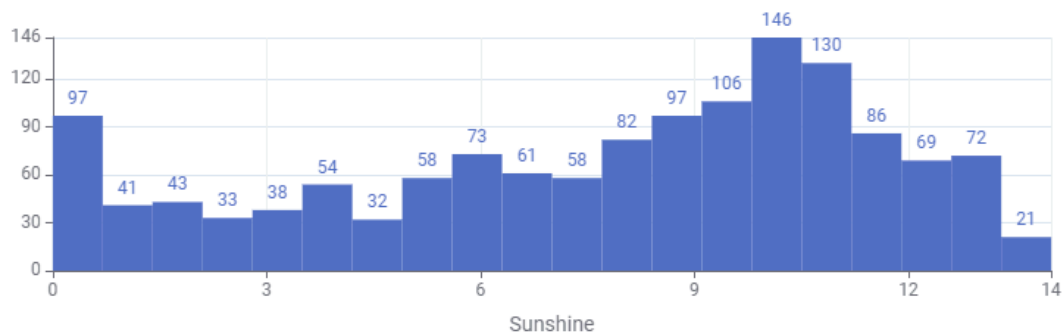


Figure 20. Histogram for attribute Sunshine

The histogram shows sunshine hours with most days getting between 5 and 10.7 hours of sunshine. A few days have 0 sunshine, indicating cloudy days (not on the histogram).

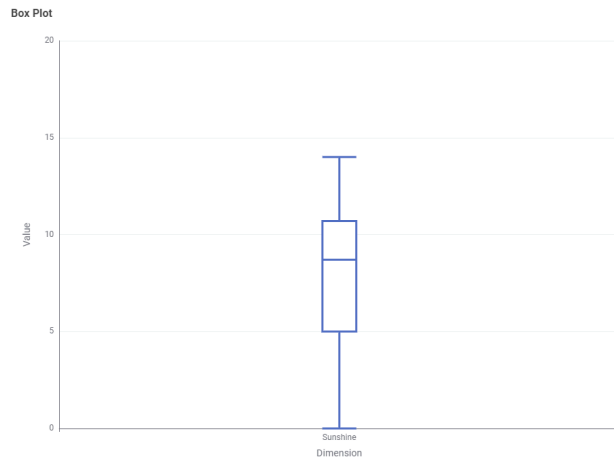


Figure 21. Box Plot for attribute Sunshine

The box plot shows a median of 8.7 hours of sunshine, with some outliers indicating days with no sunshine at all.

Attribute: WindGustSpeed

Statistics	Value
Minimum Wind Speed	7
Maximum Wind Speed	109
Mean Wind Speed	40.14
Median Wind Speed	10.686
Mode of Wind Speed	35
Variance	188.131
Standard Deviation	13.716
25th Percentile (Q1)	31
75th Percentile (Q3)	48
Skewness and Kurtosis	(0.865, 1.494)

Histogram

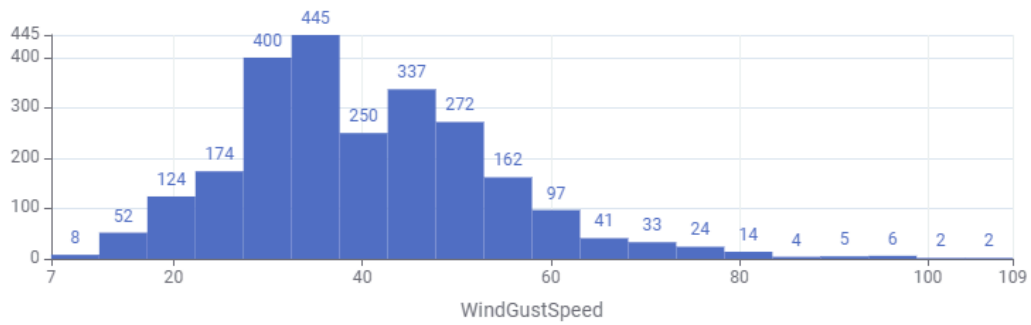


Figure 22. Histogram for attribute WindGustSpeed

The histogram shows the distribution of wind gust speeds, with most data points falling between 31 and 48 km/h. There are a few instances of very high wind speeds above 100 km/h.

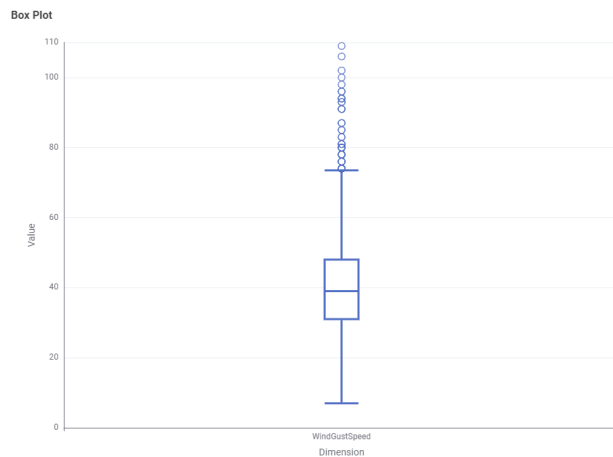


Figure 23. Box Plot for attribute WindGustSpeed

The box plot highlights the spread of wind gust speeds, with the median at 40.14 km/h. Outliers indicate days with exceptionally high wind speeds.

Attribute: WindSpeed9am

Statistics	Value
Minimum Wind Speed	0
Maximum Wind Speed	57
Mean Wind Speed	14

Median Wind Speed	13
Mode of Wind Speed	9
Variance	82.499
Standard Deviation	9.083
25th Percentile (Q1)	7
75th Percentile (Q3)	19
Skewness and Kurtosis	(0.71, 0.566)

Histogram

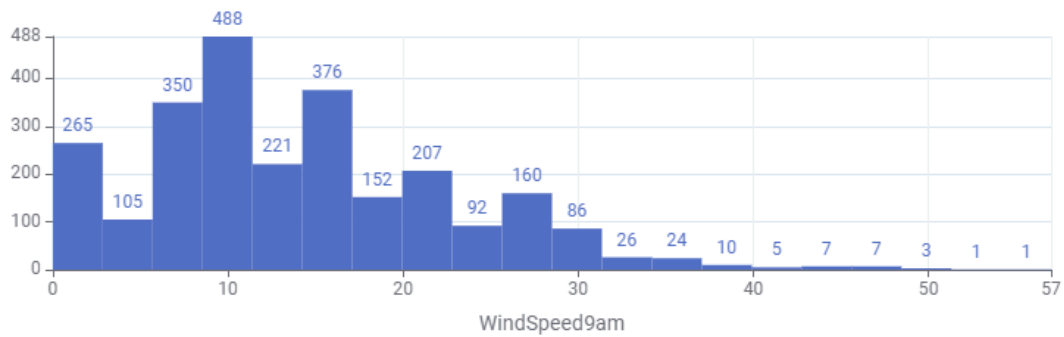


Figure 24. Histogram for attribute WindSpeed9am

Wind speed at 9am shows most values between 7 and 19 km/h, with a few calm days having wind speeds of 0.

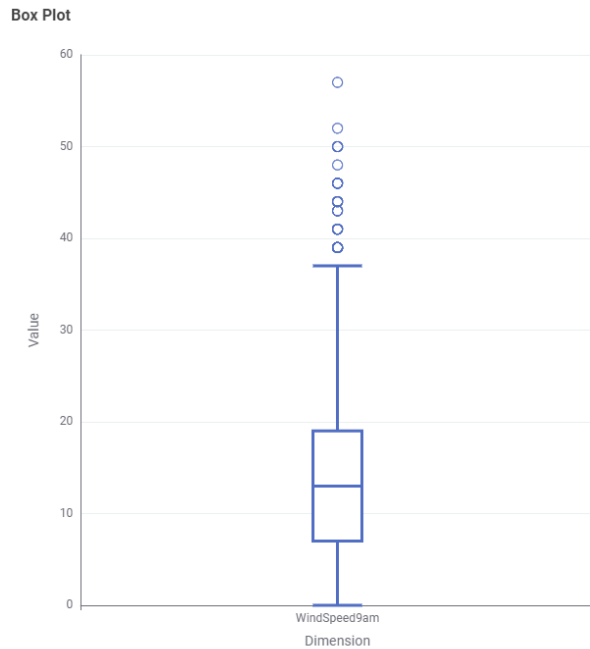


Figure 25. Box Plot for attribute WindSpeed9am

The box plot shows a median wind speed of 13 km/h at 9am, with a few outliers at higher speeds up to 57 km/h.

Attribute: WindSpeed3pm

Statistics	Value
Minimum Wind Speed	0
Maximum Wind Speed	61
Mean Wind Speed	18.57
Median Wind Speed	19
Mode of Wind Speed	15
Variance	74.035
Standard Deviation	8.604
25th Percentile (Q1)	13
75th Percentile (Q3)	24
Skewness and Kurtosis	(0.593, 0.618)

Histogram

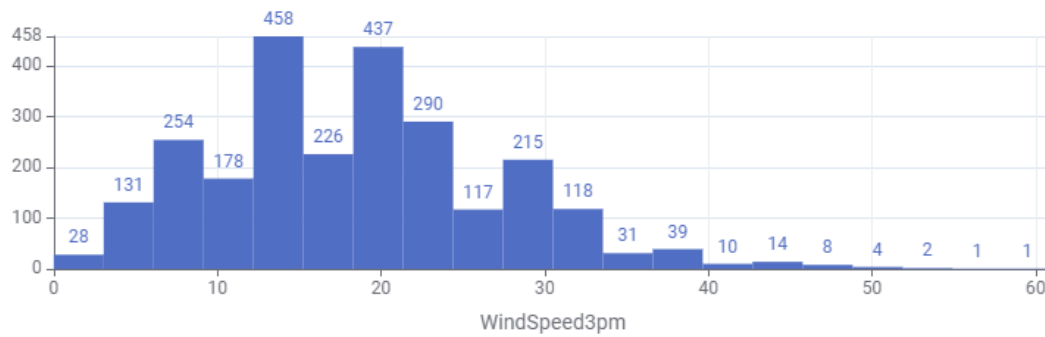


Figure 26. Histogram for attribute WindSpeed3pm

Wind speed at 3pm is generally higher, with most values falling between 13 and 24 km/h, and a few cases of wind speeds up to 61 km/h.

Box Plot

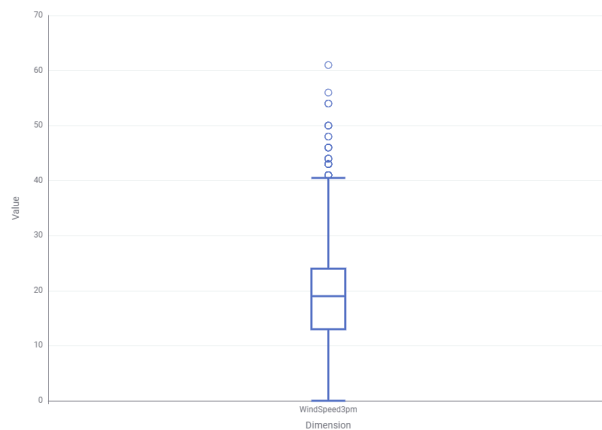


Figure 27. Box Plot for attribute WindSpeed3pm

The box plot shows the median wind speed at 19 km/h for 3pm, with higher wind speeds in the upper range but fewer outliers

Attribute: Humidity9am

Statistics	Value
Minimum Humidity	5
Maximum Humidity	100
Mean Humidity	68.767
Median Humidity	70

Mode of Humidity	99
Variance	380.489
Standard Deviation	19.506
25th Percentile (Q1)	57
75th Percentile (Q3)	83
Skewness and Kurtosis	(-0.532, 0.062)

Histogram

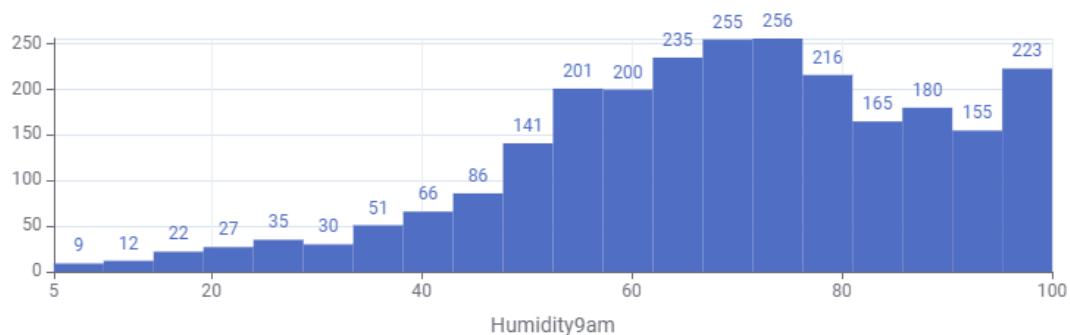


Figure 28. Histogram for attribute Humidity9am

Humidity at 9am is mostly between 57% and 83%, with a peak around 70%. A few points show extremely low humidity (5%).

Density Plot

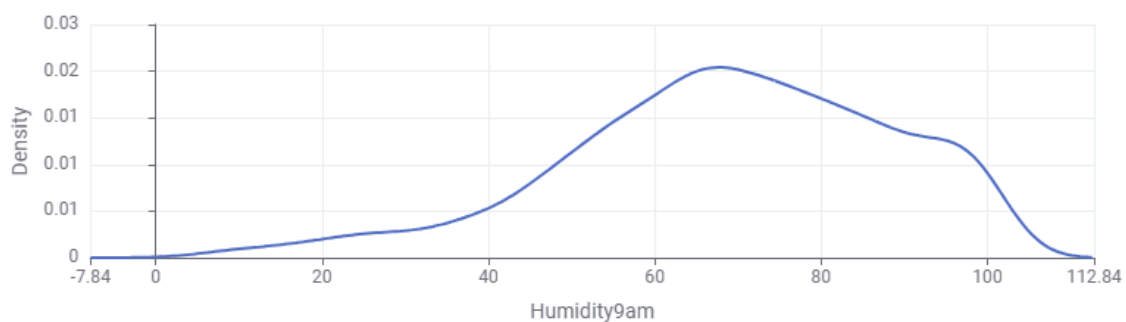


Figure 29. Density Plot for attribute Humidity9am

The density plot for humidity at 9am shows a similar distribution, with most data clustered around 70%, indicating a higher frequency of moderate humidity.

Attribute: Humidity3pm

Statistics	Value
Minimum Humidity	4
Maximum Humidity	100
Mean Humidity	51.554
Median Humidity	52
Mode of Humidity	56
Variance	457.669
Standard Deviation	21.393
25th Percentile (Q1)	35
75th Percentile (Q3)	66
Skewness and Kurtosis	(0.031, -0.594)

Histogram

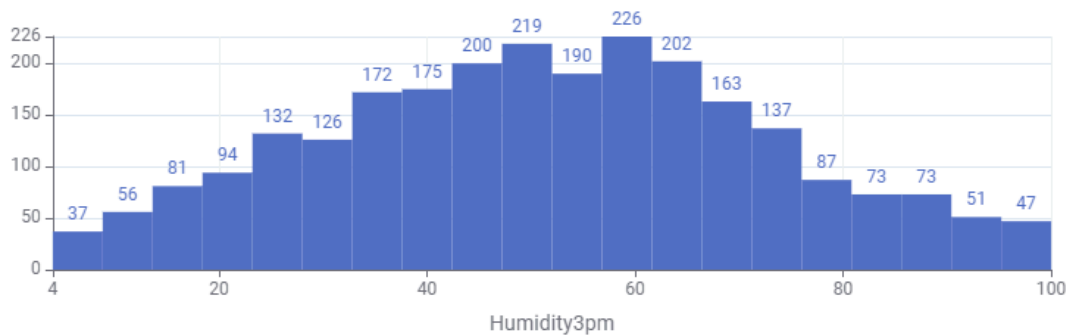


Figure 30. Histogram for attribute Humidity3pm

Humidity at 3pm is lower than in the morning, with most values between 35% and 66%. A few days show humidity close to 100%.

Density Plot

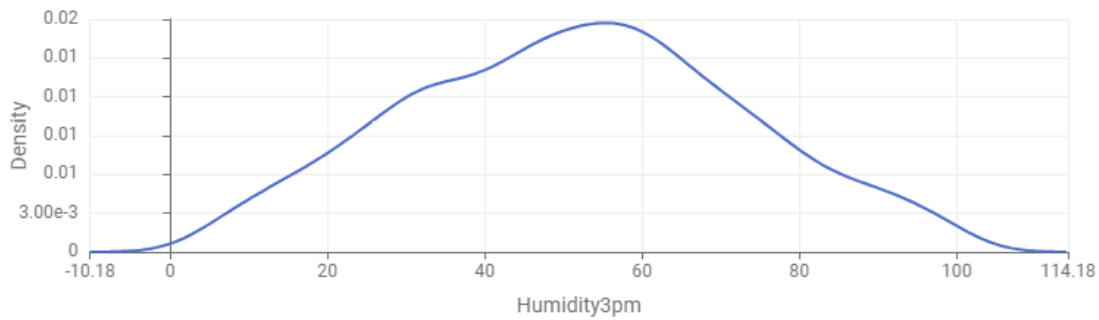


Figure 31. Density Plot for attribute Humidity3pm

The density plot for 3pm humidity shows a similar trend, with most data centered around 52%, reflecting a moderate humidity level in the afternoons.

Attribute: Pressure9am

Statistics	Value
Minimum Pressure	991.8
Maximum Pressure	1037.3
Mean Pressure	1017.698
Median Pressure	1017.7
Mode of Pressure	1016.2
Variance	49.914
Standard Deviation	7.065
25th Percentile (Q1)	1012.7
75th Percentile (Q3)	1022.5
Skewness and Kurtosis	(-0.024, 0.031)

Histogram

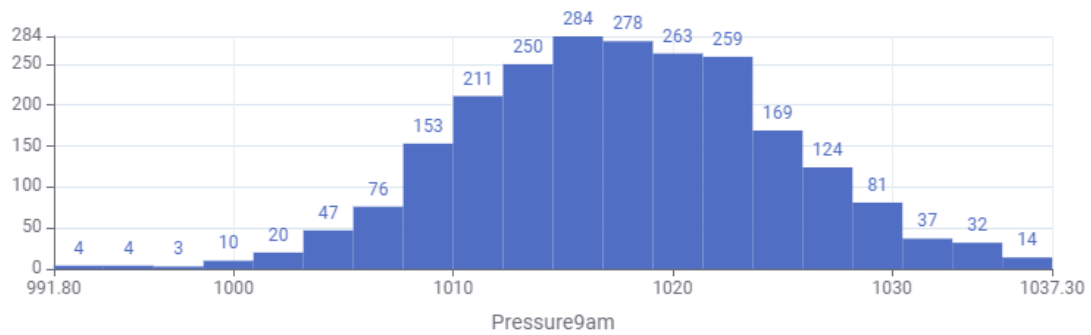


Figure 32. Histogram for attribute Pressure9am

This histogram shows the distribution of atmospheric pressure at 9am. The majority of values lie between 1012.7 and 1022.5 hPa, with a peak around 1017.7 hPa, indicating stable atmospheric conditions.

Density Plot

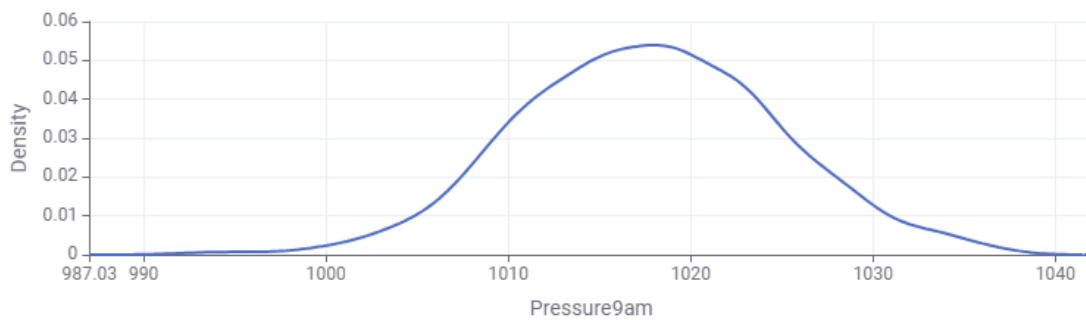


Figure 33. Density Plot for attribute Pressure9am

The density plot for Pressure9am shows a normal distribution, with a peak around 1017.7 hPa, suggesting that most days had moderate atmospheric pressure. There is minimal skewness, indicating a balanced distribution of pressure values.

Attribute: Pressure3pm

Statistics	Value
Minimum Pressure	990.3
Maximum Pressure	1036
Mean Pressure	1015.277
Median Pressure	1015.2
Mode of Pressure	1013.8
Variance	49.312
Standard Deviation	7.022
25th Percentile (Q1)	1010.2
75th Percentile (Q3)	1020.1
Skewness and Kurtosis	(0.014, -0.051)

Histogram

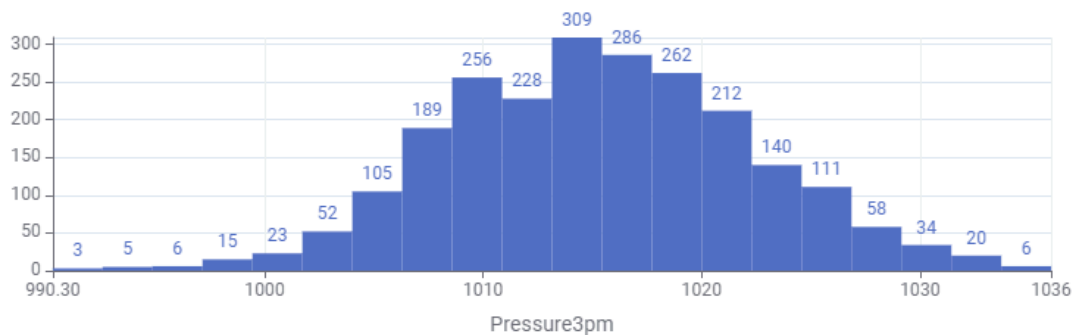


Figure 34. Histogram for attribute Pressure9am

Similar to Pressure9am, the histogram for Pressure3pm shows most values between 1010.2 and 1020.1 hPa. The peak value is slightly lower at 1015.2 hPa, reflecting minor pressure fluctuations throughout the day.

Density Plot

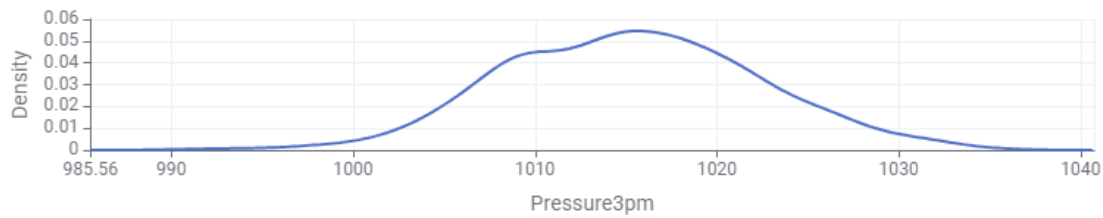


Figure 35. Density Plot for attribute Pressure3pm

The density plot for Pressure3pm displays a bell-shaped curve, with a concentration of values around 1015.2 hPa. There are no significant outliers, showing that afternoon pressure closely mirrors the morning pattern.

Attribute: Cloud9am

Statistics	Value
Minimum	0
Maximum	9
Mean	4.4
Median	5
Mode	7
Variance	8.205
Standard Deviation	2.864
25th Percentile (Q1)	1
75th Percentile (Q3)	7
Skewness and Kurtosis	(-0.186, -1.547)

Density Plot

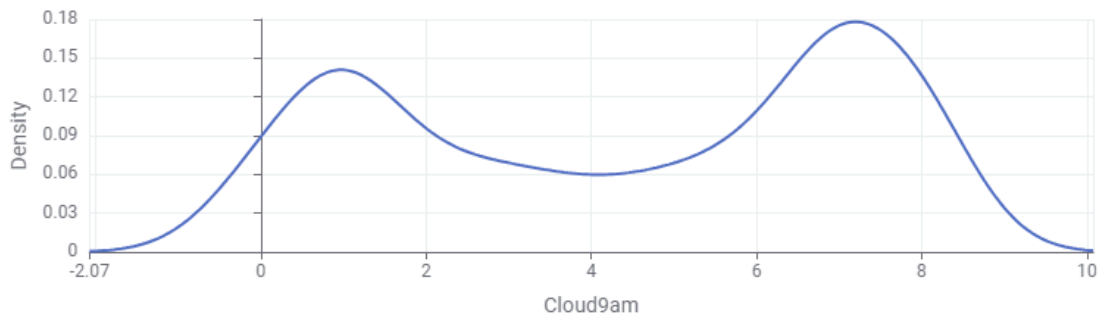


Figure 36. Density Plot for attribute Cloud9am

The cloud cover at 9am shows a fairly balanced distribution between low and high cloudiness. The peak density is around 5 octas, indicating that moderate cloud cover is common in the morning. There are few instances of completely clear skies or overcast conditions.

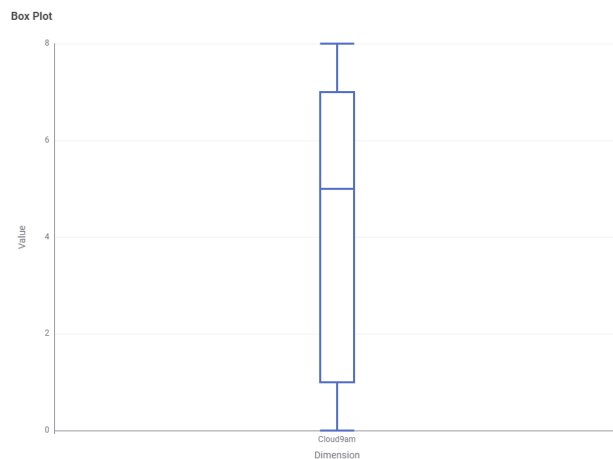


Figure 37. Box Plot for attribute Cloud9am

The box plot for Cloud9am shows that the median cloud cover is 5 octas. The interquartile range (IQR) spans from 1 to 7 octas, with few outliers, suggesting stable morning cloud conditions. A negative skew indicates a tendency toward lower cloud cover.

Attribute: Cloud3pm

Statistics	Value
Minimum	0
Maximum	8
Mean	4.574

Median	5
Mode	7
Variance	7.348
Standard Deviation	2.711
25th Percentile (Q1)	2
75th Percentile (Q3)	7
Skewness and Kurtosis	(-0.262, -1.453)

Density Plot

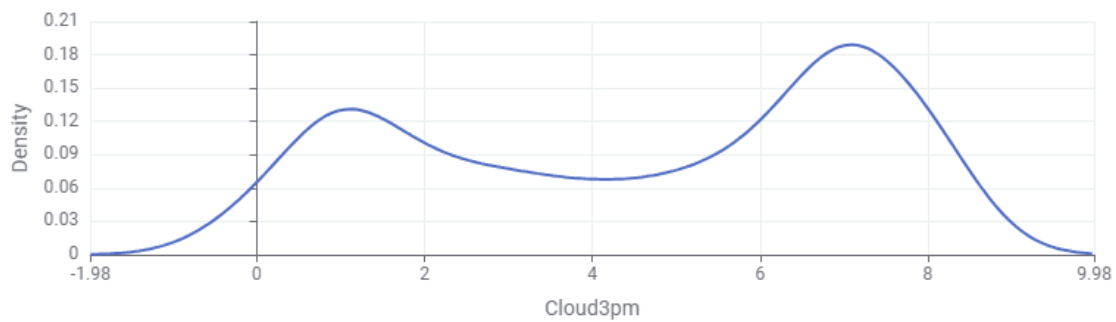


Figure 38. Density Plot for attribute Cloud3pm

Cloud cover at 3pm follows a similar pattern to 9am, with the density peaking at around 5 octas. There is a slight negative skew, showing that lower cloud cover is more common in the afternoon compared to higher values.

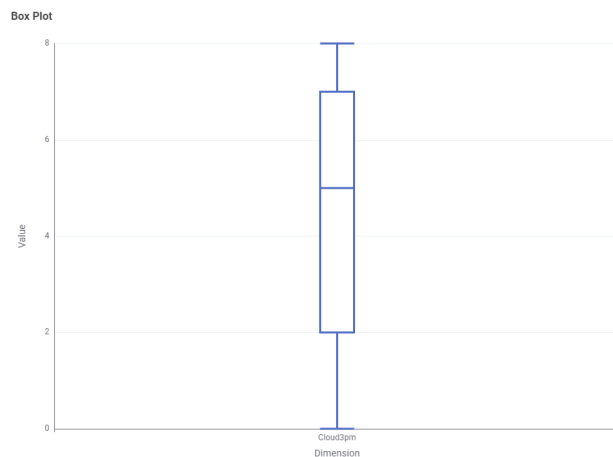


Figure 39. Box Plot for attribute Cloud3pm

The box plot for Cloud3pm shows that the median cloud cover is 5 octas. The IQR is slightly narrower than at 9am, with values ranging from 2 to 7 octas. Like the morning, there are few outliers, and the distribution suggests moderate cloud cover is typical during the afternoon.

A3. Exploration

The outlier detection has primarily been addressed in the A2 section of the assignment through the provided diagrams. In this section, I will focus on highlighting the interesting attributes and notable values associated with them.

CLUSTERING

Parallel Coordinates Plot

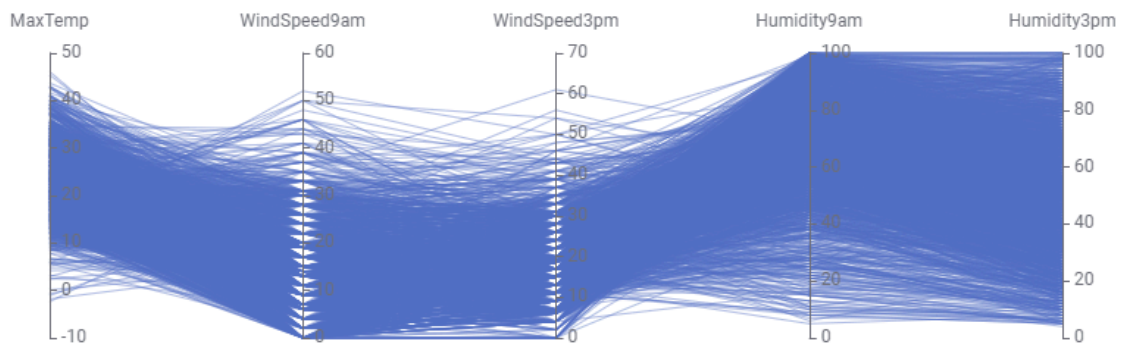


Figure 40. Clustering Analysis 1

This figure likely shows a clustering analysis of weather attributes. Clusters could indicate relationships between attributes such as temperature and humidity.

Scatter Plot

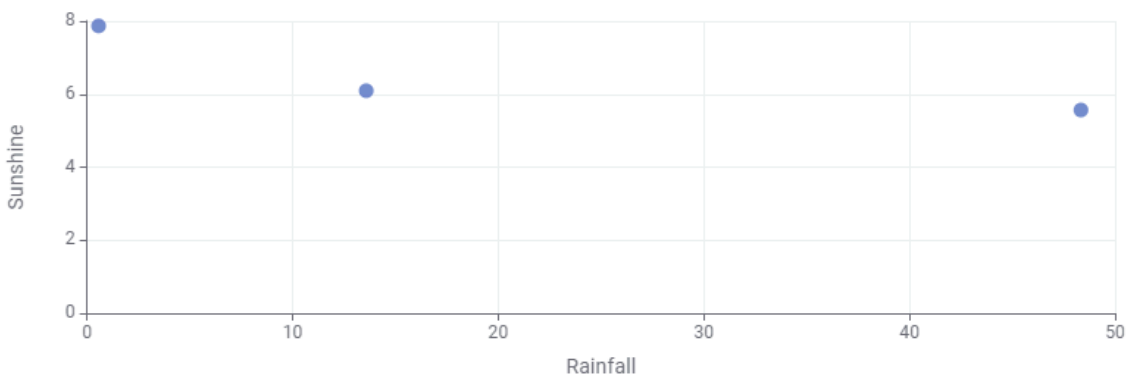


Figure 41. Relationship between Rainfall and Sunshine

The plot suggests that, in this data, there is a tendency for sunshine to decrease as rainfall increases, although it remains stable between 6-8 units in this specific case.

Scatter Plot

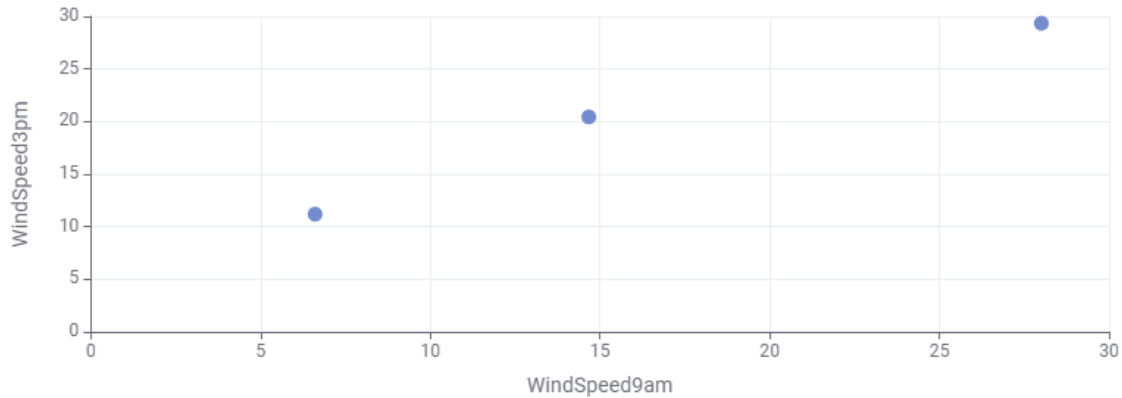


Figure 42. Scatter Plot of WindSpeed9am vs WindSpeed3pm

As seen in the plot, there is a somewhat positive correlation between these two variables. As the wind speed at 9 am increases, the wind speed at 3 pm tends to also increase.

Scatter Plot

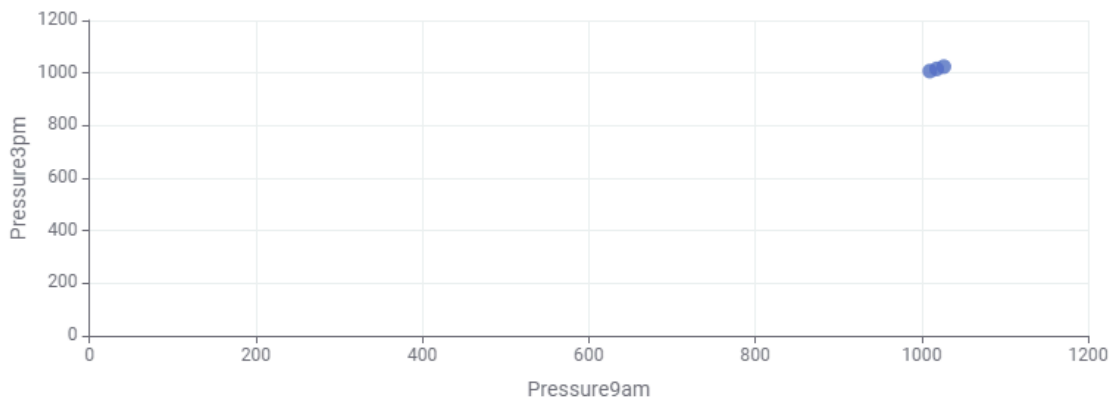


Figure 43. Comparison of Atmospheric Pressure at 9am and 3pm

The plot suggests that pressure values do not change significantly between 9am and 3pm for the data points recorded. This could reflect stable atmospheric pressure throughout the day, possibly indicating steady weather conditions.

Scatter Plot

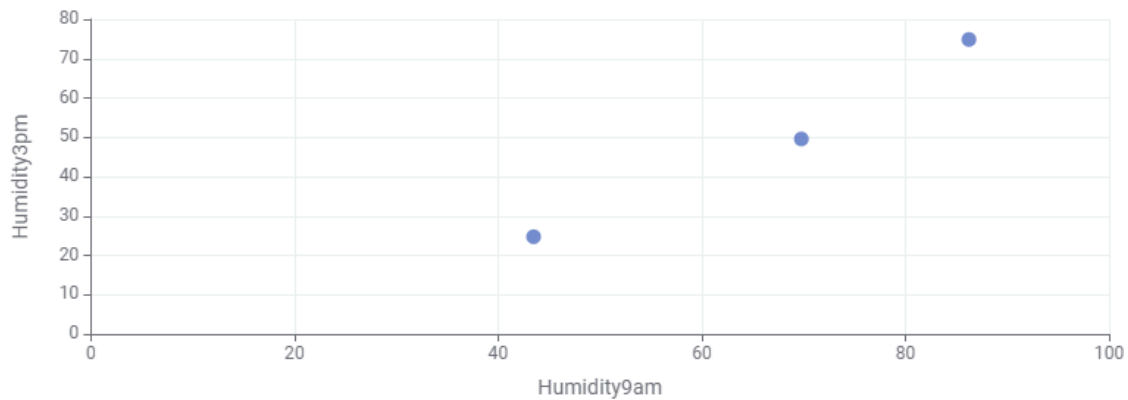


Figure 44. Comparison of Humidity Levels at 9am and 3pm

This plot suggests that the humidity levels generally decrease slightly from 9am to 3pm, though the change varies across the data points. The overall trend could indicate that as the day progresses, humidity decreases, which is typical as temperatures rise and moisture evaporates.

INTERESTING ATTRIBUTES

Scatter Plot

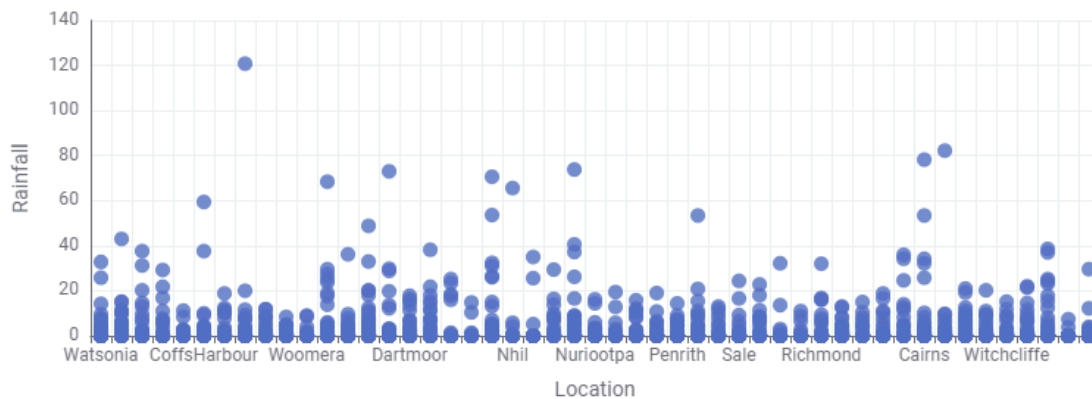


Figure 45. Rainfall Distribution Across Various Locations

The rainfall data is distributed fairly evenly across the locations, with some margins or gaps between the points, suggesting that while rainfall varies between different locations, the distribution is somewhat balanced.

Scatter Plot

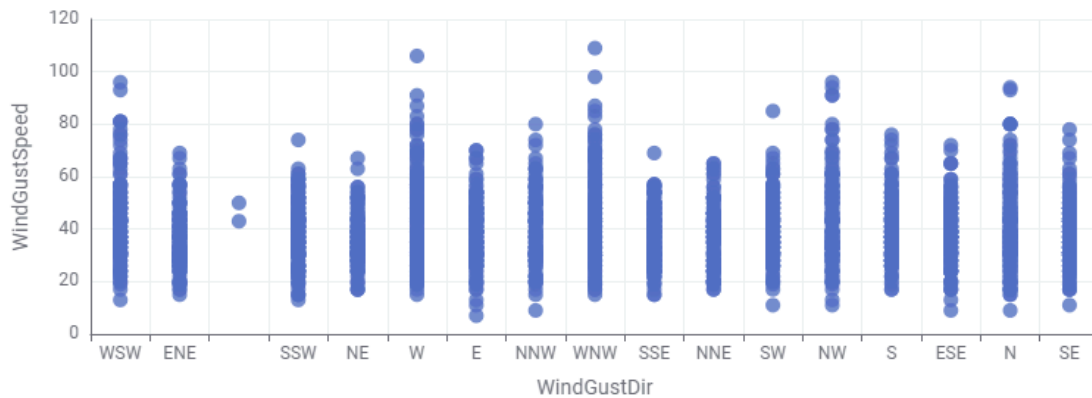


Figure 46. Wind Gust Speed Across Different Wind Gust Directions

The data suggests that wind gust speeds are relatively consistent across different wind gust directions. There are no major clusters or outliers, implying a balanced distribution between the direction from which the wind is blowing and the speed of the gusts.

Scatter Plot

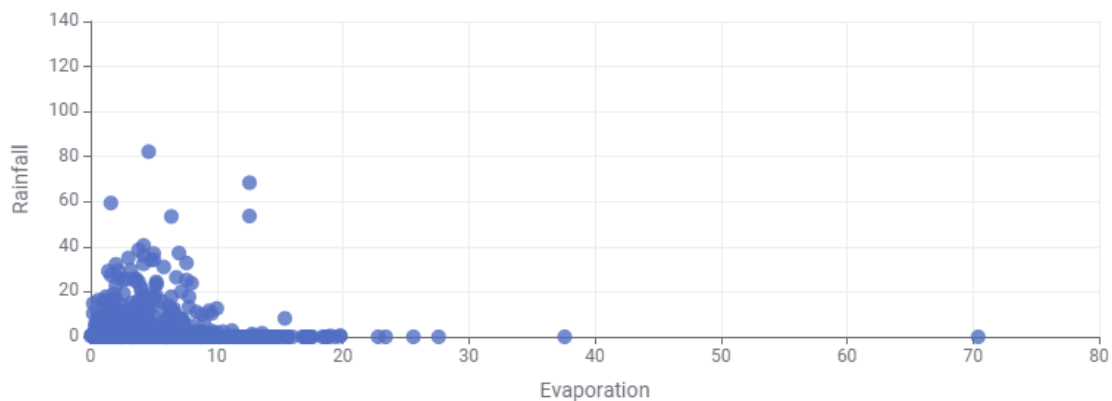


Figure 47. "Relationship Between Evaporation and Rainfall

Most of the data points are clustered to the left of the x-axis, near 0, indicating that for the majority of the cases, low evaporation levels are associated with varying amounts of rainfall.

B. Data Processing

B1. Binning Techniques

The number of bins I have decided to go with is 5. Given the characteristics of the rainfall data, this choice is justified by several key factors. The data range spans from a minimum value of 0 to a maximum value of 120.8. The dataset exhibits a significant concentration of zero values, as indicated by the mode, median, and 25th percentile (Q1) all being 0. Additionally, the 75th percentile (Q3) is only 0.8, highlighting that most of the data points are clustered at the lower end of the range.

The mean rainfall is 2.241, with a variance of 51.225 and a standard deviation of 7.157, suggesting that there is considerable variability in the data. To effectively capture this variability and provide meaningful insights, it is essential to use a sufficient number of bins.

To calculate the range for each bin, I used the following calculation:

$$(120.8-0)/5=24.16$$

Thus, the range for each bin is 24.16. This approach ensures that the bins are appropriately sized to reflect the distribution of the data, allowing for a clearer understanding of the rainfall patterns.

1. Equi-width Binning

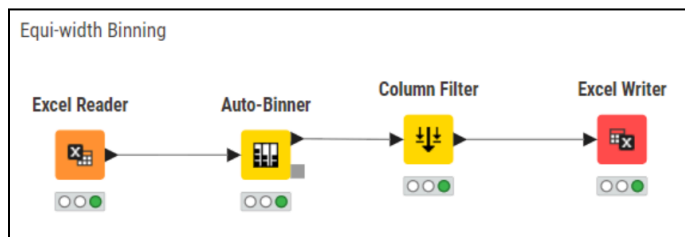


Figure 48. Steps taken to perform equi-width binning within KNIME

This figure illustrates how rainfall data is divided into equal-width bins, where each bin covers an equal range of values.

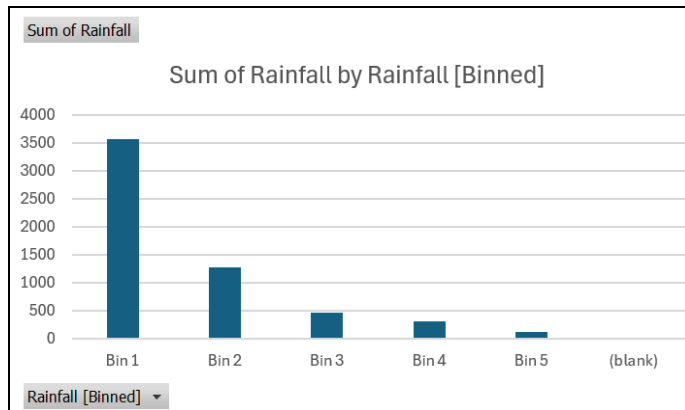


Figure 49. Histogram of Sum of Rainfall by Rainfall [Binned]

This histogram shows the distribution of rainfall data after it has been binned. The number of days with specific ranges of rainfall values is displayed

Rows: 2618 Columns: 7								
Table Statistics								
#	RowID	Rainfall Number (dou...	Rainfall [L... String	Bin 1 Number (inte...	Bin 2 Number (inte...	Bin 3 Number (inte...	Bin 4 Number (inte...	Bin 5 Number (inte...
1	Row0	14.2	Bin 1	1	0	0	0	0
2	Row1	0	Bin 1	1	0	0	0	0
3	Row2	2.8	Bin 1	1	0	0	0	0
4	Row3	0	Bin 1	1	0	0	0	0
5	Row4	0	Bin 1	1	0	0	0	0
6	Row5	0	Bin 1	1	0	0	0	0
7	Row6	0	Bin 1	1	0	0	0	0
8	Row7	120.8	Bin 5	0	0	0	0	1
9	Row8	8	Bin 1	1	0	0	0	0
10	Row9	5	Bin 1	1	0	0	0	0
11	Row10	8.6	Bin 1	1	0	0	0	0
12	Row11	0	Bin 1	1	0	0	0	0
13	Row12	17.6	Bin 1	1	0	0	0	0
14	Row13	0	Bin 1	1	0	0	0	0

Figure 50. One to Many Node for Binned Rainfall

This figure shows how the "One to Many" node was used to transform the binned rainfall data into separate columns. Each bin is represented as a new column, with a binary value indicating whether a day falls into that particular bin.

2. Equi-depth Binning

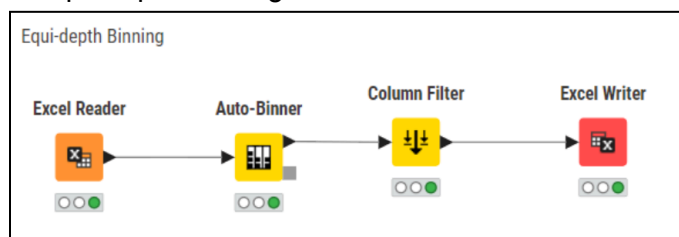


Figure 51. Steps taken to perform equi-depth binning within KNIME

This figure outlines the steps for performing equi-depth binning, where bins are created such that each contains the same number of data points.

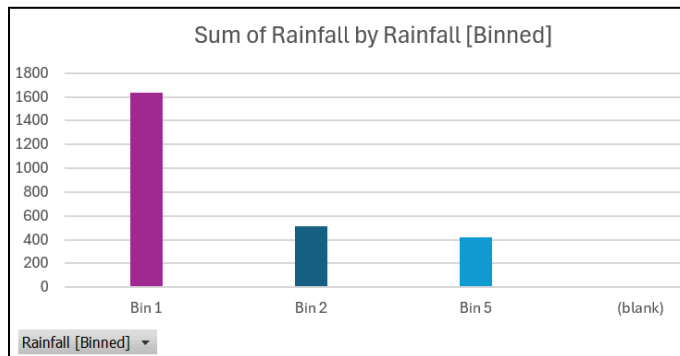


Figure 52. Histogram of Sum of Rainfall by Rainfall [Binned]

This histogram shows rainfall data after equi-depth binning.

Rows: 2618 | Columns: 7

Table Statistics

#	RowID	Rainfall Number (double)	Rainfall String	Bin 1 Number (integer)	Bin 2 Number (integer)	Bin 3 Number (integer)	Bin 4 Number (integer)	Bin 5 Number (integer)
1	Row0	14.2	Bin 5	0	0	0	0	1
2	Row1	0	Bin 1	1	0	0	0	0
3	Row2	2.8	Bin 5	0	0	0	0	1
4	Row3	0	Bin 1	1	0	0	0	0
5	Row4	0	Bin 1	1	0	0	0	0
6	Row5	0	Bin 1	1	0	0	0	0
7	Row6	0	Bin 1	1	0	0	0	0
8	Row7	120.8	Bin 5	0	0	0	0	1
9	Row8	8	Bin 5	0	0	0	0	1
10	Row9	5	Bin 5	0	0	0	0	1
11	Row10	8.6	Bin 5	0	0	0	0	1
12	Row11	0	Bin 1	1	0	0	0	0
13	Row12	17.6	Bin 5	0	0	0	0	1
14	Row13	0	Bin 1	1	0	0	0	0

Figure 53. One to Many Node for Equi-depth Binning

Like Figure 50, this figure shows how the "One to Many" node transforms equi-depth binned rainfall data into binary columns, with each bin assigned its own column.

KNIME Steps:

1. Excel Reader: Reads the dataset from an Excel file.
2. Auto-Binner: Automatically categorises the Rainfall attribute into bins.
3. Column Filter: Filters out unnecessary columns, keeping only the relevant ones.
4. Excel Writer: Writes the processed data back to an Excel file

B2. Normalisation

1. Normalisation (min-max)

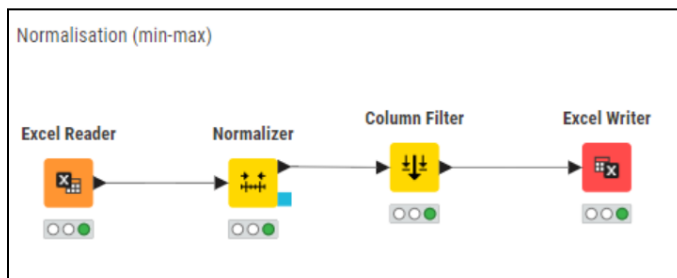


Figure 54. Min-Max Normalisation

This figure shows the result of applying min-max normalization to the attribute MaxTemp, transforming the values to a range between 0 and 1.

Rows: 2618 Columns: 1			
Table Statistics			
☐	#	RowID	MaxTemp Number (double)
☐	1	Row0	0.425
☐	2	Row1	0.485
☐	3	Row2	0.462
☐	4	Row3	0.606
☐	5	Row4	0.693
☐	6	Row5	0.656
☐	7	Row6	0.291
☐	8	Row7	0.503
☐	9	Row8	0.476
☐	10	Row9	0.462

Figure 55. Column Filter Node for Min-Max Normalisation

The result transforms the original values into the normalised range

KNIME Steps:

1. Excel Reader: Reads the dataset from an Excel file.
2. Normaliser: Applies min-max normalisation to transform the 'MaxTemp' values onto the range [0.0-1.0].
3. Column Filter: Filters out unnecessary columns, keeping only the relevant ones.
4. Excel Writer: Writes the processed data back to an Excel file

2. Normalisation (z-score)

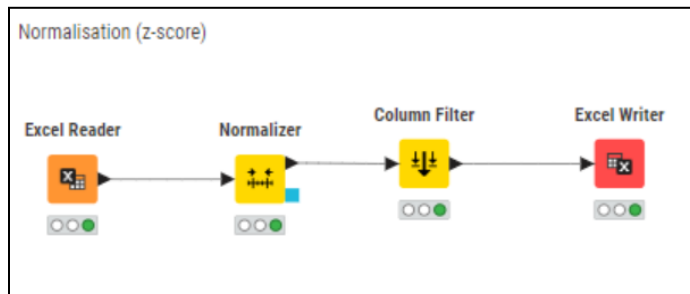


Figure 56. Z-score Normalisation

This figure presents the result of z-score normalisation, which standardises the data by subtracting the mean and dividing by the standard deviation.

Rows: 2618 Columns: 1		Table Statistics	
☐	#	RowID	MaxTemp Number (double)
☐	1	Row0	-0.69
☐	2	Row1	-0.29
☐	3	Row2	-0.442
☐	4	Row3	0.524
☐	5	Row4	1.104
☐	6	Row5	0.855
☐	7	Row6	-1.587
☐	8	Row7	-0.166
☐	9	Row8	-0.345
☐	10	Row9	-0.442
☐	11	Row10	-0.456

Figure 57. Column Filter Node for Z-score Normalisation

This figure details the results for applying z-score normalisation within KNIME

KNIME Steps:

1. Excel Reader: Reads the dataset from an Excel file.
2. Normaliser: Applies z-score normalisation to transform the 'MaxTemp' values
3. Column Filter: Filters out unnecessary columns, keeping only the relevant ones.
4. Excel Writer: Writes the processed data back to an Excel file

B3. Discretisation

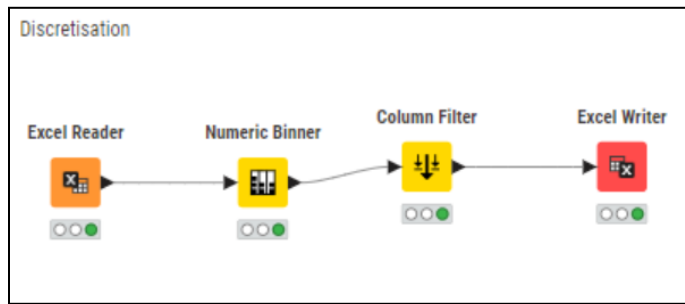


Figure 58. Steps for Discretisation of WindSpeed3pm

This figure details the steps for applying discretisation within KNIME

KNIME steps:

1. Excel Reader: Reads the dataset from an Excel file.
2. Numeric Binner: Applies rules to categorise 'WindSpeed3pm' into 'Slow Wind', 'Medium Wind', 'Fast Wind', and 'Very Fast Wind'.
3. Column Filter: Adds the categorised wind speeds as a new column in the dataset.
4. Excel Writer: Writes the updated dataset back to an Excel file.

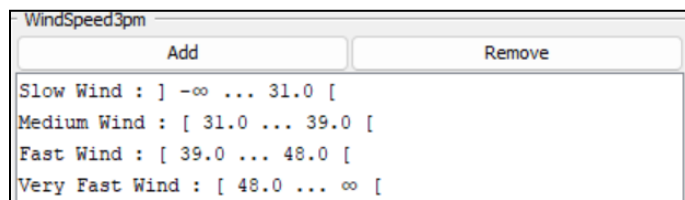


Figure 59. WindSpeed3pm Discretisation Categories

The WindSpeed3pm attribute has been categorised using 4 different category:

1. Slow Wind = 0 – 31
2. Medium Wind = 31 – 39
3. Fast Wind = 39 – 48
4. Very Fast Wind = 48 +

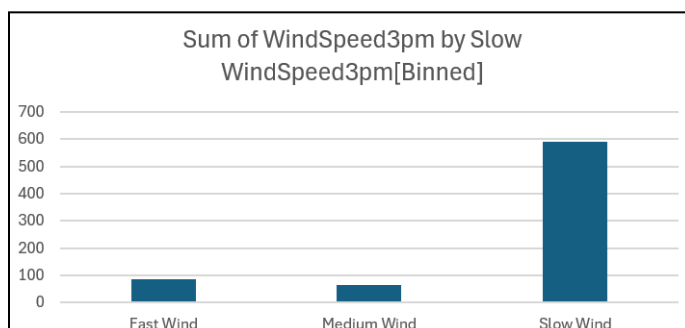


Figure 60. Histogram for the Discretisation of WindDir9am

Rows: 2618 | Columns: 2

Table Statistics

☐	#	RowID	WindSpeed3pm Number (integer)	WindSpeed3pm_binned String
☐	1	Row0	6	Slow Wind
☐	2	Row1	6	Slow Wind
☐	3	Row2	4	Slow Wind
☐	4	Row3	31	Medium Wind
☐	5	Row4	11	Slow Wind
☐	6	Row5	24	Slow Wind
☐	7	Row6	22	Slow Wind
☐	8	Row7	22	Slow Wind
☐	9	Row8	15	Slow Wind
☐	10	Row9	28	Slow Wind

Figure 61. Column Filter Node for Discretisation

This figure details the results for applying discretisation within KNIME

B4. Binarisation

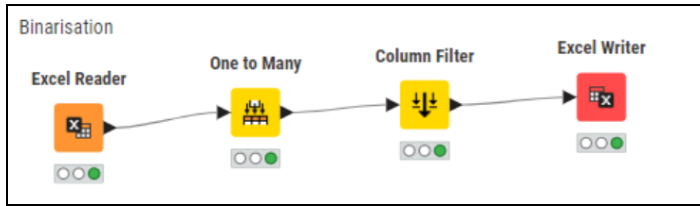


Figure 62. Binarisation of WindDir9am

This figure demonstrates the binarisation of the WindDir9am attribute, where wind directions are converted into binary values

KNIME steps:

- 1. Excel Reader: Reads the dataset from an Excel file.
- 2. One to Many: Transforms categorical data into multiple columns, with each category becoming its own column.
- 3. Column Filter: Adds the categorised wind speeds as a new column in the dataset.
- 4. Excel Writer: Writes the updated dataset back to an Excel file.

Rows: 2618 | Columns: 17

Table Statistics

☐	#	RowID	WindDir9am String	ESE Number (inte...)	S Number (inte...)	NE Number (inte...)	NNW Number (inte...)	SW Number (inte...)	SSE Number (inte...)	☐
☐	1	Row0	E	0	0	0	0	0	0	
☐	2	Row1	ESE	1	0	0	0	0	0	
☐	3	Row2	E	0	0	0	0	0	0	
☐	4	Row3	S	0	1	0	0	0	0	
☐	5	Row4	NE	0	0	1	0	0	0	
☐	6	Row5	NNW	0	0	0	1	0	0	
☐	7	Row6	NNW	0	0	0	1	0	0	
☐	8	Row7	SW	0	0	0	0	1	0	
☐	9	Row8	NE	0	0	1	0	0	0	
☐	10	Row9	NNW	0	0	0	1	0	0	

Figure 63. One to Many Node for Binarisation Results

This figure shows how the "One to Many" node was applied to binarise the WindDir9am variable to obtain the results.

C. Summary

Temperature Distribution:

- Both MinTemp and MaxTemp show typical distributions, with median values around 12°C and 22.7°C, respectively. Outliers, particularly in the lower range for MinTemp (as low as -6.8°C) and the upper range for MaxTemp (as high as 46.1°C), indicate occasional extreme weather events. These extreme values suggest temperature fluctuations that are worth further investigation to understand their underlying causes.

Rainfall:

- Rainfall distribution is heavily skewed, with most days showing little to no rainfall (median = 0 mm). Extreme values, such as the maximum rainfall of 120.8 mm on certain days, point to occasional periods of heavy rain. About 22.5% of the days recorded rainfall, with most days being dry. This skewness suggests that while heavy rainfall is rare, it can be significant when it occurs.

Wind Patterns:

- Wind direction and speed show consistent patterns. WindGustSpeed has a median of 40.14 km/h, with outliers up to 109 km/h, indicating days of strong winds.
- The most frequent wind directions at gust times are SE and S, while N is common at 9am and S at 3pm, suggesting predominant directional patterns during the day.
- Clustering of wind data could provide useful insights into regional wind behaviour and its correlation with temperature and humidity.

Parallel Coordinates Plot

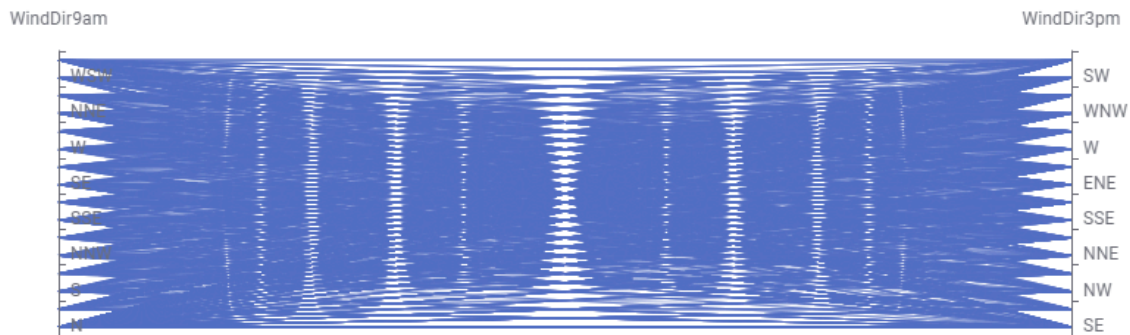


Figure 64. Wind Direction Patterns Between 9am and 3pm

Humidity:

- Humidity levels show typical daily fluctuations, with Humidity9am having a median of 70%, and Humidity3pm being lower, with a median of 52%. Extreme values include very dry mornings (as low as 5%) and highly humid days (up to 100%), which align with changes in weather patterns throughout the day.

- The data shows a moderate concentration of humidity in the mornings, with decreasing values as the day progresses, consistent with diurnal weather changes.

Sunshine and Cloud Cover:

- Sunshine hours show a reasonable distribution, with most days receiving between 5 and 10.7 hours of sunshine, and some cloudy days recording zero sunshine.
- Cloud cover follows a similar trend between morning and afternoon (9am and 3pm), with a median value of 5 octas. There is a slight skew toward lower cloud cover, indicating more frequent clear skies.

Outliers and Extreme Events:

- Outliers were identified in several attributes, including WindGustSpeed, Rainfall, and Temperature. These outliers often correspond to extreme weather events, such as storms or heatwaves, and merit further examination to understand their frequency and impact.

Clustering:

- Clustering analysis suggests that attributes like wind speed, wind direction, and temperature could be linked. For example, wind direction clusters (e.g., SE, S) correlate with temperature peaks during the day, indicating potential relationships that should be further explored.